

# EachMovie: archaeology of a first latent recommender system

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## Abstract

This retrospective presents an ongoing archaeological reconstruction of EachMovie (1995–97), believed to be the first production-scale latent recommender system; it went live online a full decade before the Netflix Prize competition. Running on two office PCs, EachMovie gave its users highly personalized movie recommendations by computing 30-dimensional latent vectors from its evolving 2.8 million-vote dataset (97.63% sparse). Its Joint Quartic Polak-Ribière iterative solver converged rapidly and used a novel “Unfilled” objective function to overcome the inherent problems of zero- or mean-filled matrices. Its Early Voting data-augmentation strategy pre-loaded and stabilized the latent space every week by importing votes from professional reviews. EachMovie worked surprisingly well, cultivating a high degree of user engagement and trust, and noted movie critic Roger Ebert called it “uncannily accurate.”

## CCS Concepts

• Information systems → Recommender systems; Collaborative filtering; • Computing methodologies → Latent variable models; Factorization methods; • Social and professional topics → History of computing.

## Keywords

Software archaeology, latent factors, low-rank matrix approximation, matrix factorization, sparsity, conjugate-gradient, cold start.

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## 1 “If you can name some movies you’ve liked ... we can name some others you might like too!”

EachMovie (1995–97) was an early research website designed, implemented, and operated at Digital Equipment Corporation’s Systems Research Center (DEC SRC) that recommended movies to its users using latent-factor models of their tastes. These models were calculated in real time by a persistent *Predictor* process as users added/changed/removed votes, and by a background *Solver* process. EachMovie’s goal was to create highly personalized movie

recommendations for its users. Its calculations were quite involved for its time, but the results obtained were the state of the art.

I was the originator of and principal researcher for EachMovie from 1995 until leaving DEC in 1997. My SRC colleague Paul McJones took over until the project ended later in 1997.

Due to DEC’s slow-motion collapse at the time, the lessons of EachMovie were never collected and published, leaving it literally prehistoric.<sup>1</sup> Mostly fragmentary artifacts remain [4] [15], and some of their current owners are unclear. Even so, I have recently received a private cache of the original (noisy) datasets and assorted (inconsistent) source-file snapshots and others, and have used these in the ongoing reconstruction of EachMovie presented here. This work was aided by an AI-based analysis (using Google’s Gemini 3 AI assistant [11]) to provide a firm foundation for this retrospective presentation.<sup>2</sup> I must emphasize, of course, that many portions of this paper are inescapably based solely on my own imperfect personal recollections, thirty years on.

## 1.1 Research contributions

This paper follows a strict archaeological reconstruction/audit of EachMovie’s design/implementation/operation. The research contributions are threefold:

- Reconstruction of the/a first latent recommender system. I document the architectural choices that let a pair of office PCs<sup>3</sup> compute 30-dimensional latent models from an evolving online dataset of 1,628 movies, 72,916 users, and 2,811,983 votes (97.63% sparse).
- Implementation analysis of the Solver. I present the Joint Quartic error polynomial used with the Polak-Ribière conjugate gradient (CG) solver to drive EachMovie’s latent-factor search much more efficiently than the alternatives.
- Early Voting and expert seeding. I describe experience with the vote-import strategy using professional reviews to pre-load and stabilize EachMovie’s latent geometries.

## 2 The *mise-en-scène*

One of my chosen research directions in 1995 was, broadly stated, to discover better ways to “find things,” either on the Internet (as with the contemporaneous research on AltaVista [1] at DEC SRC and sister labs in Palo Alto) or in the real world (as with movies).<sup>4</sup>

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<sup>1</sup>The final EachMovie dataset was subsequently available to the research community until 2004 [12] and was widely used.

<sup>2</sup>AI assistance let me keep a cautious ethical/legal distance from the recovered artifacts. I am still working to obtain the permissions to share these more broadly.

<sup>3</sup>With 133 MHz Pentium processors, 64 MiB / 96 MiB RAM, and EIDE hard drives spinning at maybe 5400 rpm.

<sup>4</sup>EachMovie predated movies on the Internet. It included every movie in theatrical release in the US and Canada during its operation, as manually entered from websites

During the design and implementation of EachMovie, I was often ignorant of related prior work and proceeded by engineering intuition to “invent” numerous useful techniques, many of which (it turns out) already existed. In this paper I use modern terminology to describe EachMovie, and highlight a few truly novel innovations.

My work was technically inspired by earlier Information Retrieval (IR) research at Bell Labs. Susan Dumais’s work on *Latent Semantic Indexing* (LSI) [6] inferred the semantics of documents from their textual contents. LSI constructed a matrix of *documents*  $\times$  *words*, counting the occurrences of each word in each document. Since the overall structure of this matrix was defined by relatively few latent factors, as expected from the *effective rank* assumption, a low-rank approximation could capture (some/much of) the semantic “signal” of the documents’ content. LSI used standard *Singular Value Decomposition* (SVD) [8] to factor its matrix into  $U\Sigma V^T$ , then used the principal components for document retrieval.

### 3 “To infinity, and beyond!”

I decided EachMovie’s user-movie vote matrix would also have low effective rank,<sup>5</sup> as with LSI, but devised a simpler rank- $K$  approximation  $U_K V_K^T$ , where  $U_K$  and  $V_K$  held the *non-orthonormal* user and movie latent vectors. (EachMovie chose  $K = 30$ .)<sup>6</sup> The dot-products of user vectors and movie vectors gave the predicted ratings.<sup>7</sup>

It was my deliberate choice to build and operate EachMovie atop a mass-market PC hardware/software platform<sup>8</sup>, to help explore alternative platforms for SRC’s research during a period of technological flux. While the theoretical foundations of EachMovie drew from LSI’s *Big Math*, its implementation was strongly influenced by its own *Small Iron*.

EachMovie’s webserver called out to a persistent *Predictor* as users added/changed/removed votes; it cached all movie vectors in RAM and streamed user votes from a shared database. A background *Solver* iterated over the database to update the latent vectors, pinging the Predictor every so often to trigger a refresh (e.g., every 90 minutes / thirty Solver iterations).

EachMovie’s vote matrices were of course very sparse.<sup>9</sup> While it is easy to “eliminate” missing votes by zero-filling their empty matrix entries, EachMovie used a novel *Unfilled* objective function instead, where the objective function minimized the least-squares

error exclusively for *observed* votes:

$$E = \sum_{(i,j) \in \Omega} W_{ij} \underbrace{(\text{Vote}_{ij} - \vec{u}_i \cdot \vec{v}_j)^2}_{\text{observed-vote error}} + \lambda \underbrace{\left( \sum_i \|\vec{u}_i\|^2 + \sum_j \|\vec{v}_j\|^2 \right)}_{L_2 \text{ regularization}}$$

Here,  $\Omega$  represents the observed-vote indices, and  $\vec{u}$  and  $\vec{v}$  are rows of  $U_K$  and  $V_K$ . While this equation differs structurally from other formulations,<sup>10</sup> it had significant advantages, as shown in Section 7.1. Ignoring missing votes protected niche signals from the “gravity of the unknown” inherent in artificially zero- or mean-filled SVD.

User votes were input on a discrete scale of 0–5 stars but held internally on a scale of 0.0–1.0. The  $W_{ij}$  vote weightings accommodated EachMovie’s novel **Sounds Awful** votes.<sup>11</sup> EachMovie set the  $L_2$  regularization constant  $\lambda = 2.0$  to eliminate numerical instability and avoid scaling to just denormals/infinities/NaNs.<sup>12</sup>

This is a **Joint Quartic** polynomial over the users’ and movies’ latent factors, used to adjust all the latent factors at once, as guided by the global gradients. This differs from the quadratic-only *Alternating Least Squares* (ALS) approach, which zigzags separately across the rows and columns. Born of my own naïve design purity, this happy accident let the CG Solver navigate the 2,236,320-scalar landscape<sup>13</sup> about 20 $\times$  faster than ALS, also as shown in Section 7.1.

EachMovie’s Solver used the *Polak-Ribière* (PR) variant of CG descent [19] instead of *Fletcher-Reeves* (FR). FR can stagnate due to imperfect numerical results, while PR’s ability to reset if/when blown off course was crucial to EachMovie, as also shown below.

Local minima were not a problem for EachMovie. Modern optimization theory [2] suggests that typical higher-dimensional landscapes are dominated by saddle points rather than true local minima, helping gradient-based solvers avoid being trapped forever.

EachMovie was built around a classic positive-feedback loop: *Votes In*  $\rightarrow$  *Predictions Out*  $\rightarrow$  *Votes In*. EachMovie gained new users from word of mouth, and from some unexpected free advertising on AltaVista’s landing page. Since many users of the era seemed to enjoy intensive online interactions, the EachMovie website design let them keep voting on and on for as many movies as desired.

### 4 Early Voting and expert seeding

New movies came out on Fridays. Their professional reviews were published on Thursdays, one day earlier.

To better support EachMovie’s positive-feedback loop, a manual **Early Voting** strategy imported new reviews each Thursday. Professional reviewers became persistent *pretend* users in EachMovie, with each new review imported as a pretend vote. This let EachMovie compute the new movies’ latent vectors in time to recommend them to real users for Friday viewing.

and local newspapers, and from weekly photocopies from *Variety*. Its DB also included many earlier movies in theatrical rerelease or in video release.

<sup>5</sup>The *actual* rank of a large  $|U| \times |V|$  matrix is  $\min(|U|, |V|)$  with high probability.

<sup>6</sup>Too large a choice of  $K$  can lead to overfitting. SVD produces a *unique* solution, but EachMovie’s latent vectors were free to rotate/scale/shear in  $K$ -space, presumably reducing overfitting by allowing redundancies in the latent factors.

<sup>7</sup>Low-rank approximations let EachMovie treat similar movies as rough semantic synonyms. Just as LSI pulled *film* and *cinema* in a common direction, EachMovie brought together movies with related fans—and users with related tastes—performing a similar semantic decomposition. Our team briefly explored extending EachMovie to use higher-order tensors to model additional user inputs regarding multi-way interactions. One application was to be a *Comments* feature to allow votes on other users’ movie comments, with EachMovie choosing which to display; this was abandoned after deployment attracted toxic content (“so gay”) that EachMovie could not yet adequately moderate automatically. Another feature, never begun, would have allowed user-authored personal movie axes (e.g., date-night vs thriller), giving those users multiple votes per movie along these multiple axes.

<sup>8</sup>Mass-market computers of that era were about 1000 $\times$  slower than today’s.

<sup>9</sup>In LSI, most matrix entries are *zero* since most documents don’t contain most words (and vice versa). In EachMovie, most entries are *missing* since most users haven’t voted on most movies—which is very different.

<sup>10</sup>EachMovie’s votes form a sparse bipartite graph, not a complete one. EachMovie Maxim™: If your data don’t fit some algorithm, change algorithms, not your data.

<sup>11</sup>EachMovie recorded “Sounds Awful” votes as 0 but with a reduced weighting of 0.2 to model vague negative signals—often based on trailers or hearsay—without overly distorting the surrounding latent geometries.

<sup>12</sup>Setting  $\lambda = 2.0$  was the equivalent of  $\lambda = 0.08$  on the external 0–5 scale. The recovered source code sometimes names  $\lambda$  as the constant hundred, providing archaeological evidence that  $\lambda = 100$  was once tried too, perhaps with votes on EachMovie’s earlier external scale of  $F^-$  to  $A^+$  that might have been manipulated on that same external scale. (Unfortunately, no revelatory comments/logs/documents remain.)

<sup>13</sup>EachMovie reached 72,916 users and 1,628 movies, with  $K = 30$ , and the Solver was solving over  $(|U| + |V|) \times K$  scalars.

Early Voting was part of a larger *expert seeding* data-augmentation strategy. Importing listicles of The Top 101 Movies Of All Time™ provided not just extra bonus votes, but also strengthened a semantic “spine” for the 30-D latent space. Even so, though professional reviewers can provide dense signals across a wide breadth of movies, it remains my belief that real users could do as useful a job of structuring the latent space *in the long run*, meaning that these manual imports were needed only to get/keep the ball rolling. I hope to test this hypothesis in future work.

#### 4.1 Roger Ebert and a cautionary tale

When Roger Ebert called EachMovie “uncannily accurate” [7], it seemed like a major win,<sup>14</sup> but I soon worried it was all an unfortunate misunderstanding due to an unanticipated **Mirror** effect.

As described above, Early Voting and expert seeding imported numerous professional reviews into EachMovie, including several entered from the *pretend* Roger Ebert. When the *real* Roger Ebert then tried EachMovie IRL and found it so “uncannily accurate,” maybe he had simply glanced into an accidental Mirror and liked what he saw. That is, EachMovie might have been primed so as to adapt so quickly to the real Roger Ebert’s own test votes, then provide such compelling recommendations right back, as to trick him into writing a better review than if EachMovie had not already imported a small yet representative body of his own reviews.

Or maybe not, of course, and it may be impossible to know so long after the fact. Still, imagining such practices *did* somehow trigger too good a review for EachMovie, I offer my belated and posthumous apologies to the real Roger Ebert.

### 5 User experiences

In any case, EachMovie *did* often seem uncannily accurate. Users were invited to write in, and many did:<sup>15</sup>

- *Just wanted to slip out a quick note—this is a greatly enjoyable site! It has led us (my wife and myself) to some really great movie choices that we otherwise would have overlooked, but I guess that was the idea, wasn’t it :)*
- *Well, last night I went out to the video store and rented a couple of movies that figured highly on my “www.eachmovie.com” prediction list. Amazing! One (Like Water for Chocolate) was an absolute masterpiece ... The other (Carrington) I’ve rated as a 4! ... You’ve got my attention! Let me congratulate you!*
- *I’ve been using EachMovie for about a month now and, I have to admit, you are pretty good in your predictions. Not all the time, of course you’re not God, but much of the time you’re close.*
- *I’ve been using it for two months, and it’s eerily good at predicting what I’ll like ... One of my favorite games with EachMovie is to enter a bogus rating, just to see how it affects the predictions; when I revert to my true opinion, EachMovie recalculates all its predictions instantly.*

<sup>14</sup>I even designed self-congratulatory bumper stickers [3] that our team mailed free to users on request.

<sup>15</sup>Many of our respondents had never seen an automated recommender system before.

I often repeated similar one-off experiments myself—add a single test user to EachMovie, vote on a single movie,<sup>16</sup> and see what happens—and the resulting predictive insights could be startling. Consider, for example, Jim Carrey, the popular/polarizing/prolific comic actor. As soon as such a test user voted up a single Jim Carrey movie, like *The Mask*, EachMovie immediately recommended other Jim Carrey movies: *Ace Ventura: Pet Detective*, *Ace Ventura: When Nature Calls*, *The Cable Guy*, *Dumb and Dumber*, and so on. This impressed me since EachMovie contained no “Jim Carrey” metadata, meaning the Jim Carrey effect was due *solely* to latent factors.

EachMovie cultivated a high degree of user engagement. I was myself a particularly enthusiastic EachMovie user, recording a total of 465 votes over 20 months (1996–97).

### 6 Neighborhood models

While contemporaneous systems like GroupLens [16] [13] from UMinn and Ringo/HOMR/Firefly [18] from the MIT Media Lab relied on *k*-nearest neighbor (*k*-NN) heuristics, EachMovie used latent factors. Neighborhood models seemed easy to compute but structurally shallow, failing to capture the transitive affinities that latent analysis so readily identifies. Also, neighborhood user-user models would seem *far* more susceptible to the Roger Ebert Mirror effect than latent user-movie models like EachMovie’s. Interestingly, Ebert found the *k*-NN recommender MovieCritic [17]—primed with large amounts of professional criticism—to have a success rate “comparable” to EachMovie’s [7], although it is possible this was largely due to the same hypothesized Mirror effect. (Gene Siskel was not as impressed with MovieCritic [20].)

### 7 From archaeology to re-animation

The reconstruction now shifts from Dr. Henry (“Indiana”) Jones, Jr., to the methods of “Dr.” Henry Frankenstein.<sup>17</sup>

#### 7.1 “Throw the switches!”

To measure the Solver’s performance, I simulated it on my M1 MacBook Pro laptop from 2020, along with a few alternative approaches. An AI-assisted rewrite into Python made this much more practical.

As described above, EachMovie used Polak-Ribière (PR) conjugate gradient descent. To test this choice, I also ran the Fletcher-Reeves (FR) CG option. Figure 1 shows PR converging rapidly—almost 50× faster than FR—as PR’s reset mechanism lets it navigate complicated geometries much more quickly.<sup>18</sup> Interestingly, an alternating PR performs *almost* as poorly as FR.<sup>19</sup>

All the iterative solvers perform better asymptotically than standard zero-filled SVD or mean-filled SVD (a fairer modern baseline that pretends each missing vote equals the overall mean). This suggests EachMovie’s Unfilled objective function models its data better than SVD, capturing more useful information.

<sup>16</sup>EachMovie’s Unfilled objective function meant these test votes did not overly damage other users’ or movies’ latent vectors.

<sup>17</sup>Yes, Indiana Jones and Herr Frankenstein were both named “Henry” [21] [24].

<sup>18</sup>These 2000 iterations would have taken over four days on EachMovie’s original PC hardware from 1996. The simulated Solver ran over 100× faster in Python on my more modern laptop from 2020, even slowed down by the various Python taxes.

<sup>19</sup>For a fair comparison, each row-column pair counts here as a single iteration.

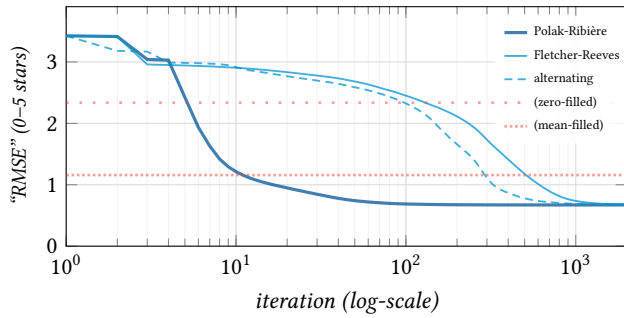


Figure 1: EachMovie’s Polak-Ribière (PR) Solver performance across  $2 \times 10^3$  iterations, vs alternatives.

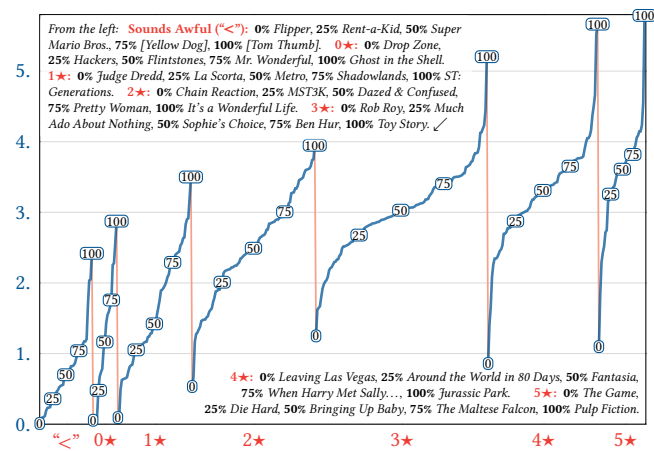


Figure 2: My 465 votes, binned L-R by value then sorted L-R by LOOCV prediction, with quartile boundaries marked.

Figure 1 is provided only to compare the speeds of the iterative solvers; its *Root-Mean-Square Error* (RMSE) values are inaccurate since none of the votes were withheld, presumably leading to overfitting. There are many better ways to compute the RMSE, of course, with varying strong points. Applying *Leave-One-Out Cross-Validation* (LOOCV) [22] to my own 465 votes (with permission) gives an RMSE of 0.9767 stars.<sup>20</sup> Figure 2 bins my votes—Sounds Awful, then 0–5 stars—plotted against their ascending LOOCV predictions, marking the movies at each quartile boundary.

To add a more user-centered measure, I also performed a top-five precision audit across 100 randomized trials—withholding a different thirty of my votes each trial—resulting in a *Closed-Set Precision@5* (P@5) [14] score of 65%.<sup>21</sup> This is a  $2.46\times$  lift compared to random chance, and a respectable value for a movie-recommender system from thirty years ago.

<sup>20</sup>Compared to 0.9568 at  $K = 20$  and 0.9499 at  $K = 25$ , so  $K = 30$  was too large.

<sup>21</sup>This audit counted only how many *withheld* 4- or 5-star movies appeared in the resulting top-five predictions. I had 121 4- and 5-star votes, about 26% of my total. My thirty votes withheld each trial were randomly selected from among *all* my votes.



Figure 3: The resurrected EachMovie, running with my own user account and its final dataset of movies and votes.

## 7.2 “It’s alive, it’s alive!”

For a final proof of concept, I also resurrected EachMovie’s webpage-generating code. Taking the Predictor’s sorted dot-products of users’ latent vectors against their unseen movies’, this code generates (Netscape-compatible) HTML for the user.<sup>22</sup> (I performed portions of this simulation by hand.) Applied to my own EachMovie account, the whole patchwork assemblage generates the webpage shown in Figure 3. This is the first time EachMovie has run since being shut down late in 1997.

## 8 Back to the future

And the end of all our exploring  
Will be to arrive where we started  
And know the place for the first time.

*Little Gidding*, T.S. Eliot, 1942

Researchers have experimented with many architectures, but modern recommender systems have largely re-converged to the latent-factor foundations pioneered by EachMovie. Even with LLMs as sophisticated encoders for Two-Tower models, their latent embeddings feed into the same dot-product geometries that EachMovie inferred on two office PCs.<sup>23</sup> EachMovie happened back when computation, cinema, and code could collide to form a purist system that can still seem “uncannily accurate,” thirty years on.

<sup>22</sup>EachMovie’s Web environment lacked CSS, cookies, JavaScript, and SSL.

<sup>23</sup>Their Small Iron is now landfill, their SiO<sub>2</sub> just dust, but the Big Math lives on.



## Acknowledgments

Many thanks to Paul McJones for his work on EachMovie at SRC, for his archival guidance, and for reviewing drafts of this paper. Other reviewers included Julian McAuley (UCSD CSE), Christine Seawell (UCI English), Jerry Kuch (UWashington CSE), and Mary-Claire van Leunen [23]. Thanks also to Mike Looney, Vanessa Doherty, Carl Waldspurger, Mark Manasse, and Lyle Ramshaw for their work on EachMovie at SRC, to the Center for Open Science for hosting the permanent EachMovie repository [5],<sup>24</sup> and to T.S. Eliot for inspiring the name “EachMovie” [9].

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<sup>24</sup> AUTHOR’S NOTE WHILE IN PUBLICATION: The EachMovie repository has moved to GitHub ([eachmovie.github.io](https://eachmovie.github.io)) and to Zenodo ([zenodo.org/communities/eachmovie/](https://zenodo.org/communities/eachmovie/)).